

Leakage-free temporal protocol

H History OD

0-35% · model input

Y Current labels

35-70% · exact regional gain

F Future test

70-100% · frozen evaluation

**Road graph
+ historical OD**

**Z0
parameter-free
bidirectional
diffusion**

**BRIDGE
frozen Z0
+ neural residual**

**BRIDGE-B
budget-aware
deployment head**

**Disjoint
region selection**

Spatially isolated
candidate regions
(Jaccard groups)

Selective CRP-style
boundary overlay
(existing exact substrate)

Exact bidirectional query
+ local endpoint access
(distance preserved)

Learning allocates offline materialization resources; online shortest-path answers remain exact.